FLUID MECHANICS

Subject Code: 23A01303

UNIT - 5: Analysis of Pipe Flow

B.Tech / JNTUA R23 / Handwritten Notes

Syllabus: Energy losses in pipelines; Darcy–Weisbach equation; minor losses in pipelines; hydraulic grade line and total energy line; concept of equivalent length – pipes in parallel and series.

1. Energy Losses in Pipelines

When a real fluid flows in a pipe, a part of its energy is dissipated as heat due to viscosity and turbulence. This is called head loss (energy loss). Total head loss in a pipeline has two parts:

Major losses (h_f)	Minor losses (h_m)
Due to friction between fluid layers and pipe wall along the full length of the pipe	Due to local disturbances: change in section, bends, valves, fittings, entry, exit
Proportional to pipe length	Occur at a point/short length
Computed by Darcy–Weisbach / Chezy / Manning formulae	Computed as $K V^2/2g$ (K = loss coefficient)
Dominant in long pipes	Can be neglected when $L/D > 1000$ (long pipes), but not in short pipes

$$h_L = h_f + h_m$$

Reynolds Number and Flow Regimes in Pipes

- $Re = \rho V D / \mu = V D / \nu$. Laminar if $Re < 2000$; transitional 2000 – 4000; turbulent if $Re > 4000$.
- Laminar flow – head loss $\propto V$; turbulent flow – head loss $\propto V^{1.75-2}$.

2. Darcy–Weisbach Equation (Major Loss)

Consider steady uniform flow through a horizontal pipe of diameter D and length L. Let p_1, p_2 be the pressures at the ends and τ_0 the average wall shear stress.

Analysis of Pipe Flow

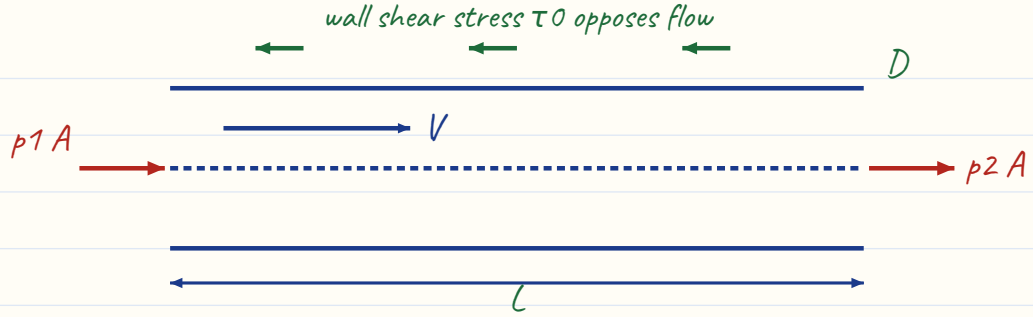


Fig 5.1: Force balance on a pipe element

Force balance in the flow direction (uniform flow – no acceleration):

$$(p_1 - p_2)(\pi D^2/4) = \tau_0 (\pi D L) \rightarrow (p_1 - p_2)/\rho g = h_f = 4\tau_0 L/(\rho g D)$$

From dimensional analysis/experiment, wall shear stress is proportional to the dynamic pressure: $\tau_0 = (f/8)\rho V^2$ where f = Darcy friction factor. Substituting:

$$h_f = f L V^2/(2 g D) \text{ (Darcy–Weisbach equation)}$$

In terms of discharge ($V = 4Q/\pi D^2$): $h_f = 8 f L Q^2/(\pi^2 g D^5)$

- h_f = loss of head due to friction (m of fluid); L = length of pipe; D = diameter; V = mean velocity; f = friction factor (dimensionless).
- Pressure drop: $\Delta p = \rho g h_f = f (L/D)(\rho V^2/2)$.
- Some books use the Fanning friction factor $f' = f/4$, giving $h_f = 4 f' L V^2/(2 g D)$ – check the definition used.
- Observation: $h_f \propto L$, $\propto V^2$ (fully rough turbulent), $\propto 1/D$ (for fixed V), $\propto Q^2/D^5$ (for fixed Q) – hence doubling the diameter reduces head loss by 32 times for the same discharge!
- Power lost in friction = $\rho g Q h_f$

Friction Factor f

Flow condition	Expression for f
Laminar (Re < 2000)	$f = 64/Re$ (Hagen–Poiseuille); independent of roughness
Turbulent, smooth pipes, 4000 < Re < 10^5	$f = 0.316 / Re^{0.25}$ (Blasius)
Turbulent, smooth pipes, any Re	$1/\sqrt{f} = 2 \log_{10}(Re\sqrt{f}) - 0.8$ (Prandtl–von Kármán)
Fully rough turbulent flow	$1/\sqrt{f} = 2 \log_{10}(D/\varepsilon) + 1.14$ (von Kármán–Nikuradse)
Transition (commercial pipes)	$1/\sqrt{f} = -2 \log_{10}[\varepsilon/(3.7 D) + 2.51/(Re\sqrt{f})]$ (Colebrook–White)

Analysis of Pipe Flow

- Hagen–Poiseuille (laminar): $h_f = 32\,\mu V L / (\rho g D^2) \rightarrow \Delta p = 32\,\mu V L/D^2$ (obtained by putting $f = 64/Re$ in Darcy's equation).
- Relative roughness ε/D: ε (absolute roughness) – commercial steel 0.045 mm; wrought iron 0.046 mm; galvanised iron 0.15 mm; cast iron 0.26 mm; concrete 0.3 – 3 mm; PVC/drawn tubing ≈ 0.0015 mm.
- In turbulent flow f depends on Re and ε/D; for large Re, f depends on ε/D only (the curves flatten).

Moody Diagram

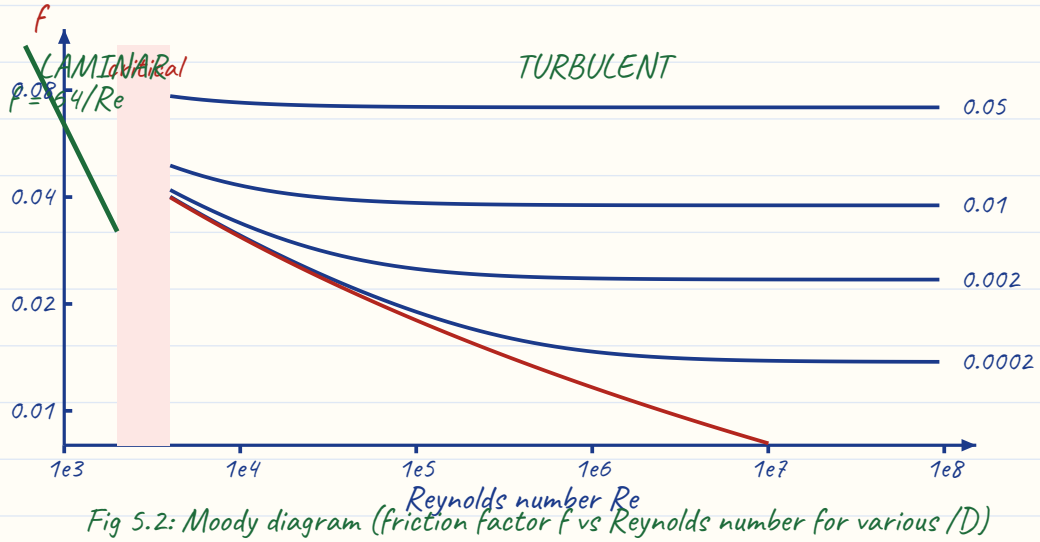


Fig 5.2: Moody diagram (friction factor f vs Reynolds number for various /D)

- Plot of f against Re (log–log) for various ε/D; it gives f for any pipe. Laminar region: straight line $f = 64/Re$. Critical zone ($2000 < Re < 4000$): uncertain. Transition: curves fall with Re. Fully rough: horizontal lines.
- Procedure: calculate Re and ε/D; read f from the chart; then $h_f = f\,L\,V^2/2g\,D$.

Other Empirical Formulae (for open and closed channels)

- Chezy's formula: $V = C\sqrt{(m\,i)}$, where m = hydraulic mean depth (hydraulic radius) = $A/P = D/4$ for a circular pipe running full, $i = h_f/L$ (slope), C = Chezy's constant. Then $C = \sqrt{(8g/f)}$.
- Manning's formula: $V = (1/N)\, m^{2/3}\, i^{1/2}$ (N = Manning roughness coefficient).
- Hazen–Williams: $V = 0.849\, C_{HW}\, R^{0.63}\, S^{0.54}$ (water supply pipes; $C_{HW} \approx 100$–140).

Example 1 – Head loss by Darcy–Weisbach

Water flows at 0.05 m^3/s through a 500 m long, 200 mm diameter pipe, $f = 0.02$. Find the head loss and the pressure drop.

Sol: $V = Q/A = 0.05/0.031416 = 1.5915$ m/s; $V^2/2g = 2.533/19.62 = 0.1291$ m

$h_f = f\,(L/D)\,V^2/2g = 0.02 \times (500/0.2) \times 0.1291 = 6.455$ m

$\Delta p = \rho g h_f = 9810 \times 6.455 = 63.3$ kPa; Power lost $= 9810 \times 0.05 \times 6.455 = 3.17$ kW

Example 2 – Laminar flow of oil

Oil ($S = 0.9$, $\mu = 0.1$ N·s/m²) flows at 1 m/s through a 50 mm pipe, 100 m long. Check the type of flow and find the head loss and pressure drop.

Sol: $Re = \rho VD/\mu = 900 \times 1 \times 0.05/0.1 = 450 < 2000 \rightarrow$ laminar. $f = 64/Re = 0.1422$

$h_f = f\,L\,V^2/(2gD) = 0.1422 \times 100 \times 1/(19.62 \times 0.05) = 14.5$ m of oil (check: $32\,\mu VL/(\rho g D^2) = 14.5$ m ✓)

$\Delta p = \rho\, g\, h_f = 900 \times 9.81 \times 14.5 = 128$ kPa

Example 2(b) – Blasius formula

Water ($\nu = 10^{-6}$ m²/s) flows at 1 m/s in a smooth 100 mm pipe of length 200 m. Find f and the head loss.

Sol: $Re = V D/\nu = 1 \times 0.1/10^{-6} = 1 \times 10\text{^}5$ (turbulent). Blasius: $f = 0.316/Re^{0.25} = 0.316/17.78 = 0.01777$

$h_f = f\,(L/D)\,V^2/2g = 0.01777 \times 2000 \times 0.05097 = 1.81$ m

3. Minor Losses in Pipelines

Minor losses arise from sudden changes in flow geometry which cause separation, eddies and mixing. They are expressed as:

$$h_m = K\,V^2 / 2g \quad (K = \text{loss coefficient, found by experiment})$$

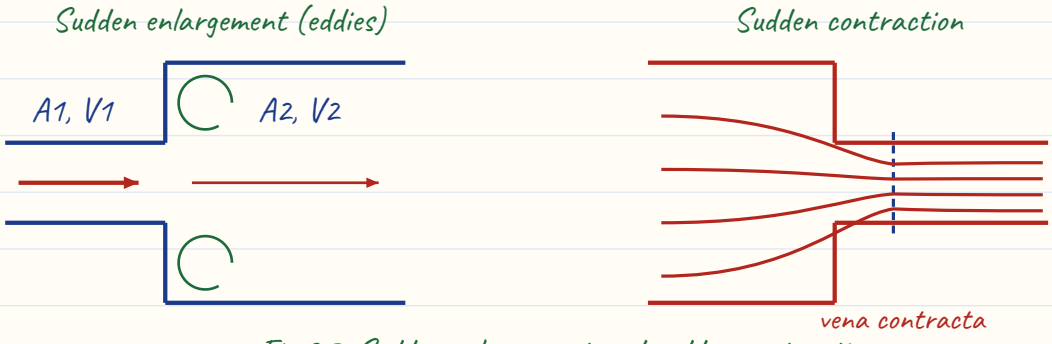


Fig 5.3: Sudden enlargement and sudden contraction

(a) Loss due to Sudden Enlargement

Applying the momentum equation between section 1 (just after the enlargement, where pressure acts on the full area A_2) and section 2, together with the energy and continuity equations (Borda–Carnot):

$$h_e = (V_1 - V_2)^2 / 2g = (V_1^2/2g)(1 - A_1/A_2)^2$$

- Pressure rise across the enlargement: $(p_2 - p_1)/\rho g = (V_1^2 - V_2^2)/2g - h_e = V_2 (V_1 - V_2)/g$.
- Exit loss from a pipe into a large reservoir ($A_2 \to \infty$, $V_2 = 0$): $h_e = V^2/2g$ ($K = 1$).

(b) Loss due to Sudden Contraction

The flow contracts to a vena contracta (area $A_c = C_c A_2$) and then expands to fill the smaller pipe; the loss is mainly the sudden-enlargement loss from the vena contracta to section 2:

$$h_c = (1/C_c - 1)^2 V_2^2/2g = K_c V_2^2/2g \ (C_c \approx 0.62 \to K_c \approx 0.375; \text{commonly } h_c = 0.5 V_2^2/2g)$$

(c) Other Minor Losses

Source of loss	Loss head	Typical K
Entrance (sharp-edged, from reservoir)	$0.5 V^2/2g$	0.5 (rounded: 0.05 – 0.2; re-entrant: 0.8)
Exit (pipe to reservoir)	$V^2/2g$	1.0
Gradual enlargement (cone)	$k (V_1 - V_2)^2/2g$	$k = 0.1 - 0.5$ depending on angle
Gradual contraction	$k V_2^2/2g$	0.02 – 0.05
Pipe bend / elbow 90°	$K V^2/2g$	0.9 (standard), 0.3 (long radius)
Elbow 45°	$K V^2/2g$	0.4
Gate valve (fully open)	$K V^2/2g$	0.15 – 0.2 (half open ≈ 5.6)
Globe valve (fully open)	$K V^2/2g$	≈ 10
Check (swing) valve	$K V^2/2g$	≈ 2.5
Loss due to an obstruction (partial blockage)	$(A/(C_c (A - a)) - 1)^2 V^2/2g$	–

Note: Total head loss of a pipeline of length L and diameter D with minor items: $h_L = (f L/D + \Sigma K) V^2/2g$.

Example 3 – Sudden enlargement

A pipe enlarges suddenly from 150 mm to 300 mm diameter. Discharge is 0.08 m^3/s. Find the head loss and the pressure rise.

Sol: $V_1 = 0.08/0.017671 = 4.527$ m/s; $V_2 = 0.08/0.070686 = 1.132$ m/s

$h_e = (V_1 - V_2)^2/2g = (3.395)^2/19.62 = 0.588$ m

$(p_2 - p_1)/\rho g = (V_1^2 - V_2^2)/2g - h_e = (20.49 - 1.28)/19.62 - 0.588 = 0.979 - 0.588 = 0.392$ m $\rightarrow \Delta p = 3.84$ kPa

Example 4 – Sudden contraction

A 300 mm pipe suddenly contracts to 150 mm; $Q = 0.05\ m^3/s$. Find the loss (use $h_c = 0.5\ V_2^2/2g$).

Sol: $V_2 = 0.05/0.017671 = 2.829$ m/s; $h_c = 0.5 \times (2.829^2/19.62) = 0.5 \times 0.408 = 0.204$ m

4. Equivalent Length (Concept)

A fitting (valve, bend, etc.) can be replaced by an equivalent length L_e of straight pipe of the same diameter that gives the same head loss. Equating minor loss and friction loss:

$$K\,V^2/2g = f\,L_e\,V^2/(2\,g\,D) \rightarrow L_e = K\,D/f$$

- The total 'virtual' pipe length $L_{total} = L + \Sigma L_e$ can be used in Darcy's equation with a single f.
- Equivalent length is often quoted in terms of diameters: $L_e/D = K/f$ (e.g., a globe valve ≈ 340 D).
- Not to be confused with the equivalent pipe (next section) – which replaces a whole compound pipe system by a single pipe.

Example 5

A globe valve ($K = 10$) is in a 100 mm pipe with $f = 0.025$. Find its equivalent length.

Sol: $L_e = K\,D/f = 10 \times 0.1/0.025 = 40$ m $(= 400\ D)$

5. Hydraulic Grade Line (HGL) and Total Energy Line (TEL)

Total Energy Line (TEL) / Energy Grade Line

A line which joins the total head at every section of a pipe: $p/\rho g + V^2/2g + z$ measured above a datum.

Hydraulic Grade Line (HGL) / Piezometric Line

A line which joins the piezometric head at every section: $p/\rho g + z$. It shows the level to which water would rise in a piezometer tube fixed to the pipe.

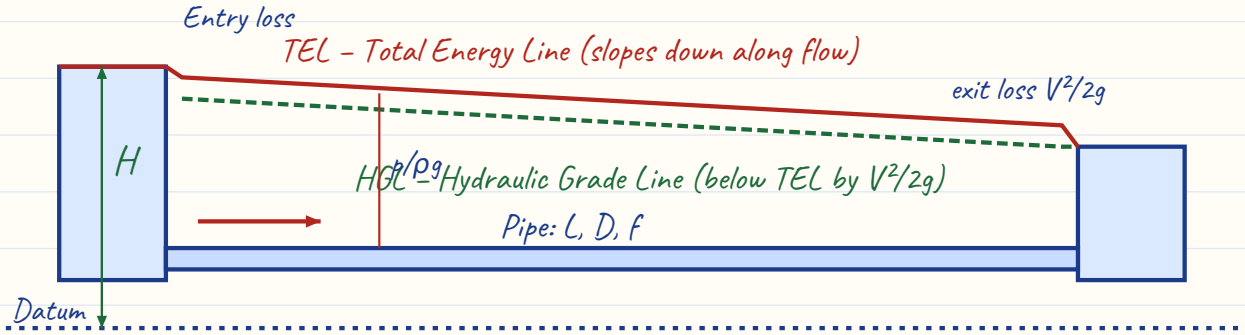


Fig 5.4: TEL and HGL for a pipe connecting two reservoirs

Properties of TEL and HGL

- TEL is always above HGL by the velocity head $V^2/2g$. For a pipe of uniform diameter the two are parallel.
- TEL always slopes downward in the direction of flow because of head losses (it rises only where a pump adds head; it falls sharply at a turbine). The slope = h_f/L = hydraulic gradient.
- HGL may rise or fall: it rises in a diverging section (velocity falls, pressure recovers) and falls in a converging section.
- At the free surface of a reservoir, TEL and HGL coincide with the water surface ($V = 0$, $p = 0$ gauge). At a free jet discharge to the atmosphere HGL passes through the centre of the outlet.
- Sudden drop in both lines occurs at local losses (entrance, valve, sudden enlargement).
- If the pipe rises above the HGL, the pressure in that portion is below atmospheric (negative gauge), e.g. at the crest of a siphon – risk of cavitation / air release. If the pipe lies below the HGL, the pressure is positive.
- Vertical distance from pipe centre-line to the HGL = pressure head at that point.
- For a long pipe where minor losses and velocity head are neglected, HGL ≈ TEL.

Example 6 – Flow between two reservoirs with minor losses

Two reservoirs with a level difference of 12 m are connected by a pipe 600 m long, D = 0.25 m, f = 0.02. Sharp-edged entrance (K = 0.5) and exit (K = 1). Find the discharge.

Sol: Energy equation between the surfaces: $H = (K_{en} + K_{ex} + f\,L/D)\,V^2/2g = (0.5 + 1 + 0.02 \times 600/0.25)\,V^2/2g = 49.5\,V^2/2g$

$V^2 = 12 \times 19.62/49.5 = 4.756 \rightarrow V = 2.181$ m/s

$Q = A\,V = 0.04909 \times 2.181 = 0.107$ m³/s (friction accounts for 48/49.5 = 97 % of the loss, so minor losses may be neglected for such a long pipe)

Example 6(b) – Tank discharging through a pipe; pressure at mid-point

A tank with water level 15 m above a horizontal pipe (D = 100 mm, L = 80 m, f = 0.02) discharges freely to the atmosphere. Sharp-edged entry (K = 0.5). Find Q and the pressure at mid-length of the pipe.

Sol: $H = V^2/2g$ (exit velocity head) + $0.5\,V^2/2g + f(L/D)\,V^2/2g = (1 + 0.5 + 0.02 \times 800)\,V^2/2g = 17.5\,V^2/2g$

$V^2 = 15 \times 19.62/17.5 = 16.82 \rightarrow V = 4.10$ m/s; $Q = 4.10 \times 0.007854 = 0.0322$ m³/s; $V^2/2g = 0.857$ m

At mid-length: $p/\rho g = 15 - V^2/2g - 0.5\,V^2/2g - f(40/0.1)\,V^2/2g = 15 - 0.857 - 0.429 - 6.857 = 6.857$ m → p = 67.3 kPa (gauge)

Application: Siphon

A siphon is a closed pipe bent over a high point (summit) that carries liquid from a higher reservoir to a lower outlet. The summit lies above the HGL, so the pressure there is below atmospheric.

- Velocity at the outlet (neglecting losses): $V = \sqrt{(2\,g\,h)}$, h = difference of level between the upper reservoir surface and the outlet.
- Pressure head at the summit: $p_s/\rho g = -(h_s + V^2/2g + \text{losses up to the summit})$, h_s = height of the summit above the upper reservoir surface.
- For satisfactory working, p_s (absolute) must stay above the vapor pressure of the liquid (otherwise vapor/air pockets break the flow – cavitation). Practical limit ≈ 7.6 m of water of vacuum at the summit.

Example 6(c) – Siphon

A siphon of 50 mm diameter discharges water to the atmosphere at a point 4 m below the upper reservoir surface. The summit is 2 m above the reservoir surface. Neglecting losses, find Q and the pressure at the summit.

Sol: $V = \sqrt{(2 g h)} = \sqrt{(2 \times 9.81 \times 4)} = 8.86$ m/s; $Q = A V = 0.0019635 \times 8.86 = 0.0174$ m^3/s

$V^2/2g = 4$ m. $p_s/\rho g = -(2 + 4) = -6$ m → $p_s = -6 \times 9810 = -58.9$ kPa (gauge) (≈ 42.4 kPa absolute – safe, above p_v).

6. Pipes in Series (Compound Pipe)

SERIES: same Q, losses add up



PARALLEL: same head loss, discharges add up

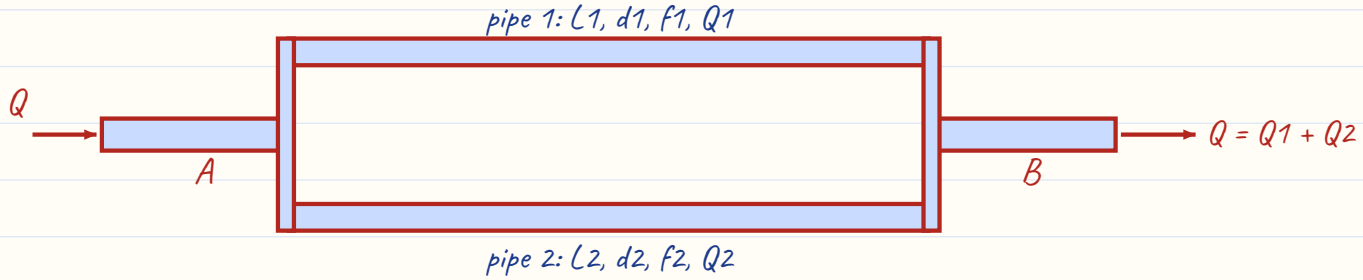


Fig 5.5: Pipes in series and pipes in parallel

Pipes of different lengths and diameters are connected end-to-end so that the whole fluid flows through each pipe successively.

- Discharge is the same in all pipes: $Q = A_1 V_1 = A_2 V_2 = A_3 V_3$.
- Total head loss = sum of individual losses (plus minor losses at junctions): $H = h_{f1} + h_{f2} + h_{f3} + h_m$
- $H = (f_1 L_1 V_1^2/2g\, d_1) + (f_2 L_2 V_2^2/2g\, d_2) + (f_3 L_3 V_3^2/2g\, d_3)$ + minor losses.

Equivalent Pipe – Dupuit's Equation

A compound pipe can be replaced by a single pipe of length $L = L_1 + L_2 + L_3$ and equivalent diameter d (or a pipe of the same diameter and equivalent length) that carries the same discharge for the same total head loss. Taking the same f and neglecting minor losses ($h_f = 8 f L Q^2/\pi^2 g d^5$):

$$L/d^5 = L_1/d_1^5 + L_2/d_2^5 + L_3/d_3^5 \quad \text{(Dupuit's equation, series)}$$

- Equivalent pipe of the same length $L = \Sigma L_i$: $d^5 = L/(\Sigma L_i/d_i^5)$. Equivalent length for the same diameter d: $L = d^5 \Sigma (L_i/d_i^5)$.
- If minor losses are included they may be expressed as equivalent lengths and added to L_i.

Example 7 – Series pipes

Two reservoirs with a level difference H = 20 m are joined by pipe 1 (L = 300 m, d = 0.3 m) in series with pipe 2 (L = 200 m, d = 0.2 m). f = 0.02 for both. Neglecting minor losses, find Q and the equivalent length for d = 0.2 m.

Sol: $H = (8f Q^2/\pi^2 g)(L_1/d_1^5 + L_2/d_2^5)$; $L_1/d_1^5 = 300/0.00243 = 123457$; $L_2/d_2^5 = 200/0.00032 = 625000$; sum = 748457

$8f/(\pi^2 g) = 0.16/96.82 = 0.001653 \rightarrow 20 = 0.001653 \times 748457 \times Q^2 = 1236.8\, Q^2 \rightarrow Q^2 = 0.01617$

$Q = 0.127\ m^3/s$. Equivalent length for d = 0.2 m: $L_e = d^5 \times 748457 = 0.00032 \times 748457 = 239.5$ m.

7. Pipes in Parallel

Two or more pipes branch from a common point A and rejoin at point B. Fluid divides among the branches and then recombines.

- Total discharge = sum of branch discharges: $Q = Q_1 + Q_2 + Q_3$.
- Head loss is the same in every branch (same pressure difference between A and B): $h_{f1} = h_{f2} = h_{f3} = h_f$
- Using $h_f = 8 f L Q^2/(\pi^2 g d^5)$: $Q_i = (\pi/(\sqrt{8}))\sqrt{(g h_f)} \cdot \sqrt{(d_i^5/(f_i L_i))} \rightarrow Q_i \propto \sqrt{(d_i^5/(f_i L_i))}$
- Equivalent pipe for a parallel system: (same length and f) $d_e^{5/2} = d_1^{5/2} + d_2^{5/2} + \ldots$ or in general $\sqrt{(d^5/L)} = \sqrt{(d_1^5/L_1)} + \sqrt{(d_2^5/L_2)} + \ldots$
- Parallel arrangement reduces the overall loss for the same Q, increasing the carrying capacity (looping of mains in water distribution).

Example 8 – Parallel pipes

A pipe system carries $Q = 0.15\ m^3/s$ through two parallel pipes each 600 m long with f = 0.02, $d_1 = 0.3$ m and $d_2 = 0.2$ m. Find the discharge in each pipe, the head loss and the equivalent diameter.

Sol: Equal h_f and same L, f: $Q_1/Q_2 = (d_1/d_2)^{5/2} = 1.5^{2.5} = 2.756$

$Q_1 = 0.15 \times 2.756/3.756 = 0.110\ m^3/s$; $Q_2 = 0.15 - 0.110 = 0.040\ m^3/s$

$h_f = 8 f L Q_1^2/(\pi^2 g d_1^5) = 8 \times 0.02 \times 600 \times 0.01212/(96.82 \times 0.00243) = 4.94$ m

Equivalent diameter: $d_e^{2.5} = 0.3^{2.5} + 0.2^{2.5} = 0.04930 + 0.01789 = 0.06718 \rightarrow d_e = 0.34$ m

Branching Pipes (Three Reservoirs Problem) – Brief

- Three reservoirs connected at a junction J: the head at J (piezometric head) is unknown. Continuity at J: $Q_1 + Q_2 = Q_3$ (flow in = flow out).
- Assume the HGL level at J, compute Q in each pipe from $h_f = f\,L\,V^2/2g\,D$, check continuity, and adjust the assumed level (trial and error) – this is the basis of pipe-network analysis (Hardy-Cross method).

8. Power Transmission through Pipes

Water at head H available at the inlet of a pipeline is used to run a turbine (or nozzle) at the other end. Head available at the outlet = $H - h_f$ (minor losses neglected).

$$\text{Power transmitted } P = \rho\, g\, Q\,(H - h_f)$$
$$\text{Efficiency } \eta = (H - h_f)/H$$

- With $h_f = 8\,f\,L\,Q^2/(\pi^2\, g\, D^5)$ (Darcy), P is zero at Q = 0 and at the maximum Q (when all head is lost in friction); it attains a maximum in between.
- Condition for maximum power: $h_f = H/3$, and the corresponding maximum efficiency = 66.7 %.
- Proof: $P \propto Q\,(H - k\,Q^2) \rightarrow dP/dQ = 0 \rightarrow H = 3\,k\,Q^2 = 3\,h_f$

Example 9 – Maximum power transmission

Water is supplied at a head of 100 m through a pipe of L = 1000 m, D = 0.2 m, f = 0.02. Find the maximum power that can be transmitted and the discharge.

Sol: For maximum power $h_f = H/3 = 33.33$ m. $h_f = f\,L\,V^2/(2gD) \rightarrow V^2 = 33.33 \times 19.62 \times 0.2/(0.02 \times 1000) = 6.54 \rightarrow V = 2.557$ m/s

$Q = 2.557 \times 0.031416 = 0.0803\ m^3/s$; $P = \rho\, g\, Q\,(H - h_f) = 9810 \times 0.0803 \times 66.67 = 52543\ W \approx 52.5\ kW$ ($\eta = 66.7$ %)

9. Quick Revision Sheet

Concept	Formula
Reynolds number	$Re = \rho V D/\mu$ (laminar < 2000, turbulent > 4000)
Darcy–Weisbach	$h_f = f L V^2/(2gD) = 8 f L Q^2/(\pi^2 g D^5)$
Laminar friction factor	$f = 64/Re$; $h_f = 32 \mu V L/(\rho g D^2)$
Blasius (turbulent smooth)	$f = 0.316/Re^{0.25}$
Wall shear stress	$\tau_0 = f \rho V^2/8$
Sudden enlargement	$h_e = (V_1 - V_2)^2/2g$
Sudden contraction	$h_e = 0.5 V_2^2/2g$ (or $(1/C_c - 1)^2 V_2^2/2g$)
Entrance / exit	$0.5 V^2/2g$ / $V^2/2g$
Equivalent length	$L_e = K D / f$
Series pipes (Dupuit)	$L/d^5 = \Sigma L_i/d_i^5$; Q same; Σh_f
Parallel pipes	$Q = \Sigma Q_i$; h_f same; $d_e^{5/2} = \Sigma d_i^{5/2}$
TEL / HGL	$TEL = p/\rho g + V^2/2g + z$; $HGL = p/\rho g + z$

Important Questions

1. Derive the Darcy–Weisbach equation for head loss due to friction in pipes.
2. Explain the Moody diagram and the variation of the friction factor with Reynolds number and relative roughness.
3. Derive expressions for the head loss due to sudden enlargement and sudden contraction.
4. List the minor losses in a pipe system. What is the concept of equivalent length?
5. Define HGL and TEL. Draw them for a pipe connecting two reservoirs with a pump in between.
6. Derive Dupuit's equation for pipes in series. Obtain the equivalent diameter for pipes in parallel.

Note: Prepared as handwritten-style study notes for 23A01303 Fluid Mechanics (Unit V). Friction-factor constants and K-values are typical textbook values; use the values given in the problem.